Shortest vector to the intersection of hyperplanes

Consider a set C which is the intersection of two hyperplanes:

$$C = \left\{ x \in \mathbf{R}^n \mid \sum_{i=1}^{n} x_i = 10, \;\; 2x_1 + 3x_2 - 4x_n = 6. \right\}$$

Find the vector x that lies on C and having the shortest distance from the origin.

Write computer function that takes n as the input, and return x and the shortest distance.

Input Format

- an integer n, $10 \leq n \leq 100$

Output Format

- $x \in \mathbf{R}^n$: a n-length array
- `shortest_distance` as a scalar

Implementation

Goal: Fill in the following function:

```
import numpy as np

def shortest_vector2hyperplane(n):

  return x, shortest_dist
```